MIL-C-85122(AS)
8 January 1980
SUPERSEDING
(See Section 6)

MILITARY SPECIFICATION

FOR

CARTRIDGES, 20 MM

MARK 100 SERIES

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification defines assembly, performance and test requirements for MARK 100 series 20 millimeter (MM) cartridges.

1.2 Classification. Cartridges covered by this specification are as follows:

(a) High Pressure Test (HPT), MARK 101 (DL 2605724).

(b) Low Pressure Test (LPT), MARK 102 (DL 2605725).

(c) Dummy, MARK 103 (DL 2605723).

(d) Target Practice (TP), MARK 105 (DL 2519699).

(e) High Explosive Incendiary (HEI), MARK 106 (DL 2519684).

(f) Armor Piercing Incendiary (API), MARK 107 (DL 2519700).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1305

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(g) Armor Piercing Tracer (APT), MARK 108 (DL 2519701).

(h) Firing Circuit Test (FCT), MARK 109 (DL 2051583).

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-250/4	Aluminum Alloy 2024, Plate and Sheet.

MILITARY

MIL-L-296	Lacquer, Purple.
MIL-P-1394	Primer, Electric, M52A3B1.
MIL-C-18759	Cartridge Case, Steel, 20-Millimeter MARK 5 MOD 0.
MIL-I-45607	Inspection Equipment, Supply and Maintenance, for Ordnance.
MIL-P-81994	Propellant for 20 MM Ammunition.
MIL-P-85121	Projectiles, 20 MM MARK 100 Series.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.

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MIL-STD-1167	Ammunition Data Card.
MIL-STD-1168	Ammunition Lot Numbering.
MIL-STD-1695	Environments, Working, Minimum Standards for.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

DL 2051583	Firing Circuit Test (FCT), MARK 109.
DL 2519684	High Explosive Incendiary (HEI), MARK 106.
DL 2519699	Target Practice (TP), MARK 105.
DL 2519700	Armor Piercing Incendiary (API), MARK 107.
DL 2519701	Armor Piercing Tracer (APT), MARK 108.
DL 2605723	Dummy, MARK 103.
DL 2605724	High Pressure Test (HPT), MARK 101.
DL 2605725	Low Pressure Test (LPT), MARK 102.

Naval Sea Systems Command
(Code Ident 10001)

LD 256008	20 MM Ammunition Packaging and Package Marking, List of Drawings, Sub-Assemblies and Parts.
1314850	20 MM Cartridge MARK 5 MOD 0 Assembled Ammunition Gage.
SK 679810	Barrel, Copper Pressure Gauge Test, MARK 100 Series, 20 MM Ammunition.
SK 679811	Barrel, Electric Gauge Test, MARK 100 Series, 20 MM Ammunition.

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PUBLICATIONS

MILITARY

Naval Air Systems Command
(Code Ident 30003)

AD-1324	Range Firing Procedures for MARK 100 Series 20 Millimeter Naval Aircraft Gun Ammunition.

Naval Sea Systems Command
(Code Ident 53711)

OP 2165 (Volume 1)	Navy Transportation Safety Handbook for Hazardous Materials.

Naval Supply Systems Command

NAVSUP Publication 505/MCO P4030.19	Packaging and Handling of Dangerous Materials for Transportation by Military Aircraft.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations

49 CFR Parts 171-178 and 180-190	Transportation.

Coast Guard Regulations

CG-108	Rules and Regulations for Military Explosives and Hazardous Munitions.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

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3. REQUIREMENTS.

3.1 General. Individual requirements for the cartridge shall be as specified herein, on the applicable detailed specification, and the applicable drawing. (See 6.2.1 for MARK and MOD ordered.)

3.1.1 Major components. The cartridge shall consist of the following major components:

(a) Cartridge case

(b) Electric primer

(c) Propellant

(d) Projectile assembly

3.2 First article. Unless otherwise specified in the contract or purchase order, the contractor shall furnish 1,250 sample' cartridges for first article-inspection and approval (see 4.4 and 6.2.1). The first article sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the first article sample shall be at the risk of the contractor. The samples shall be delivered to the testing activity designated in the contract or purchase order (see 6.2.1).

3.3 Conformance to documents. The cartridge major components and parts shall be manufactured, tested, inspected, and assembled in accordance with the requirements, processes, and procedures of the documents listed on the applicable Data List (DL) (see 1.2). When a drawing or a specification for a part invokes by reference the provisions of a military specification, conformance to the provisions of the referenced specification is required. Conformance is defined as not only conformance to the stated performance and product characteristics, but also compliance with all quality assurance provisions, including first article and quality conformance testing in accordance with the documents listed in the applicable detail specification.

3.4 Loading and assembly.

3.4.1 Primers and cases. Primers that have met the requirements of MIL-P-1394 shall be inserted and crimped into cartridge cases meeting the requirements of MIL-C-18759 as shown on the applicable drawings. Graphite or lubricants shall not be used to facilitate primer insertion.

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3.4.1.1 Primer sealing. Each cartridge case primer shall be waterproofed by application of purple lacquer (MIL-L-296), to the joint between the cartridge case and primer. Purple lacquer shall be visible after application, and the 0.140-inch-diameter at the center of the primer button shall remain free of lacquer.

3.4.1.2 Primer resistance. The electrical resistance, between the primer contact button and the cartridge case, shall be within 1 kilohm minimum and 1.2 megohms maximum.

3.4.2 Propellant. The primed cartridge case shall be loaded with propellant (MIL-P-81994) conditioned to room temperature. Care shall be exercised to maintain the loading room humidity at levels that will not alter the moisture content of the propellant during the loading process. The exact charge weight shall be within ±5 grains of the assigned weight for the index of propellant being used as determined by MIL-P-81994.

3.4.3 Projectile insertion. Projectile assemblies that meet the requirements of MIL-P-85121 and other specifications effected by the appropriate data lists (see l.2) shall be inserted into the cartridge case and crimped as required by the applicable drawing. The crimping process shall be controlled to ensure that the maximum pressure applied to the neck of the cartridge case shall not exceed 39,000 pounds per square inch (psi).

3.4.3.1 Projectile extraction. The minimum force required to extract the projectile from the cartridge after crimping shall be 1,300 pounds.

3.5 Performance.

3.5.1 Ballistics.

3.5.1.1 Velocity-pressure. Velocity and chamber pressure values shall be as follows:

(a) Average muzzle velocity, measured at 78 feet, shall be 3,320 ±50 feet per second (ft/sec) with a maximum standard deviation of 40 ft/sec.

(b) Average peak pressure at 70 degrees Fahrenheit (°F) shall not exceed (with the exception of HPT and LPT cartridges) 58,000 psi copper with a maximum standard deviation of 3,000 psi. (If a piezoelectric transducer is used to measure pressure, the pressure measured is converted to copper crusher gage pressure by dividing by 1.2.)

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3.5.1.2 Waterproof velocity. The average muzzle velocity, measured at 78 feet, shall be 3,320 ±75 ft/sec with a maximum standard deviation of 40 ft/sec.

3.5.1.3 Action time. The average action time shall not exceed 3.0 milliseconds (ms) and the individual action times shall not exceed 3.3 ms (see 6.5.3).

3.5.1.4 Accuracy. The average dispersion pattern for two groups of 10 cartridges, each fired at 8 vertical target placed at a distance of 500 yards shall not exceed the following:

(a)	TP	(DL 2519699)	1.39 mils
(b)	HEI	(DL 2519684)	0.95 mil
(c)	API	(DL 2519700)	0.95 mil
(d)	APT	(DL 2519701)	1.20 mils

3.5.1.5 Function and casualty. When fired at a minimum rate of 925 shots per minute (spin) from the MARK 1.2 gun mechanism and 3,800 spm minimum rate from the MARK 4 gun pod, the firing defects specified in Table I shall not be exceeded.

3.5.2 High pressure. The HPT cartridge (DL 2605724) shall provide a chamber pressure of 68,000 ±2,000 psi copper at 70°F when loaded with the type and quantity of propellant specified by the procuring activity (see 6.2.1).

3.5.3 Low pressure. The LPT cartridge (DL 2605725) shall provide a chamber pressure of 50,000 ±2,000 psi copper at 70°F when loaded with the type and quantity of propellant specified by the procuring activity (see 6.2.1).

3.5.4 Armor piercing incendiary. The API projectile shall produce an incendiary function upon target impact.

3.5.5 Armor piercing tracer. The APT projectile tracer shall burn for a minimum of 1.1 seconds without separation of either the tracer cup or tracer incendiary composition from the tracer cup.

3.5.6 High explosive incendiary. The HEI projectile shall produce a high-order detonation on target impact.

3.6 Safety. The projectile shall show no evidence of in-bore or in-flight premature detonation (HEI), premature ignition (API), or tracer cup separation (APT), within 2,500 yards of the gun.

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TABLE I. Maximum allowable firing defects per lot sample.

Defects	Maximum allowable defective units[1/]
Misfire	0
Projectile remaining in bore	0
Primer leak:	
(a) Large leak or evidence of gas erosion on base of case or bolt of gun	2
(b) Small leak or evidence of gas leakage around primer body	5
(c) Damaged or blown primer	0[2/]
Case casualty:	
(a) Circumferential rupture	
(1) Complete at crimping groove	0
(2) Partial; neck, shoulder, body, or base	0
(b) Longitudinal split	
(1) Shoulder or body	0
(2) Neck	3
(c) Flange cracks	[3/]
Failure to extract, including sheared rim	0[4/]
Failure to chamber	0
Gun stoppage (from ammunition defects not listed)	0
Separation of metal parts of projectile assembly during flight:	0
(a) Body Breakup	0
(b) Loss of nose, fuze, or windshield	0
(c) Complete rotating band separation	0
(d) Partial rotating band separation with band seat knurling imprint detectable on band	0

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TABLE I. Maximum allowable firing defects per lot sample (continued).

Defects	Maximum allowable defective units[1/]
(e) Partial rotating band separation with bands, having no evidence of band seat knurling imprint:	
(1) Fragments having lengths no greater than 0.500 inch	3
(2) Fragments having length greater than 0.500 inch	0
(3) Particles from normal band fringing or slivering	—[5/]

[1/] The number of defective units permitted pertain to the total number of cartridges fired as specified in Table II, except that primer leaks shall not be considered defects for cartridges fired in a single action.

[2/] Unseated or upset primer buttons without indication of gas leakage are not considered as defective units.

[3/] Flange cracks, as determined by visual examination, shall not be cause for rejection of the lot but shall be recorded and reported.

[4/] Prior to firing a light application of oil or other lubricants, as required, may be applied to the cartridge case.

[5/] Lot will not be penalized.

3.6.1 Drop sensitivity. HEI and API projectiles shall not deflagrate or function on impact with a steel plate at a velocity of 52 ±5 ft/sec or a drop, nose down, from a 40-foot height.

3.7 Identification and marking. The cartridges shall be identified and marked as specified on the applicable drawing.

3.8 Government-loaned property. The following gage will be Government-loaned property: 20 MM Cartridge MARK 5 MOD 0 Assembled Ammunition Gage (Drawing 1314850).

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3.9 Workmanship. The cartridges shall be assembled utilizing good workmanship practices common to the ordnance industry. Particular care shall be taken that all components are clean, free from grease, dirt, corrosion, chips and foreign material. Components with obvious defects or handling damage that could affect performance of the cartridge or test weapon shall not be assembled.

3.9.1 Special working conditons. Facilities used in the manufacture, testing, and inspection of cartridges shall meet the minimum standards for working conditions of MIL-STD-1695.

3.10 Ammunition data cards. Ammunition data cards (DD Form 1650) shall be completed in accordance with MIL-STD-1167 for each lot of cartridges (see 6.2.2). An ammunition data card shall be placed on top of the contents of each ammunition box.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Special working environment. To meet the requirements of 3.9.1, the contractor shall provide adequate facilities for the fabrication, assembly, and testing of supplies to be delivered in accordance with this specification.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

(a) First article inspections (see 4.4).

(b) Quality conformance inspections (see 4.5).

(1) Individual inspections.

(2) Sampling inspections.

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4.3 Inspection provisions and conditions. All firing inspections (tests) shall be accomplished at the testing activity specified in contract or purchase order (see 6.2.1).

4.3.1 Inspection equipment. Unless otherwise specified in the contract or purchase order (see 6.2.1), all inspection equipment shall be supplied and maintained by the contractor in accordance with MIL-I-45607.

4.4 First article inspection. The 1,250 first article samples (see 3.2) shall be subjected to the examinations and tests of Table II. Failure of any cartridge to meet the requirements specified herein shall be cause for rejection of the first article samples.

4.5 Quality conformance inspection. The quality conformance sample shall be subjected to the examinations and tests of Table II

4.5.1 Production lot. The production lot shall be not less than 20,000 cartridges and not more than 50,000 cartridges. The lot shall consist of the following:

(a) Projectiles from one lot.

(b) Cartridge cases from one lot.

(c) Primers from one lot.

(d) Propellant from one index.

(e) Tracers from one lot.

(f) Fuzes from one lot.

4.5.1.1 Mixed component lot. When the lot size has been established, all components that are left over are to be retained until a sufficient number is accumulated for a mixed lot of 20,000 to 50,000 cartridges. The accumulated leftovers of propellant shall not be used in mixed component lots unless blended by a method acceptable to the procuring activity as determined by the Government representative, with a charge determination made for its assembly into the cartridges. The blending method shall be made available for review by the Government representative. Mixed components shall not be used without prior approval of the procuring activity. The ammunition data card shall contain information on all lots of components used.

4.5.1.2 Lot identification. Each cartridge and each packed cartridge lot shall be identified in accordance with applicable drawings and MIL-STD-1168, supplemented as directed by the procuring activity (see 6.2.1).

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TABLE II. First article and quality conformance inspection.

Test or examination	Quantity	Requirement	Test
Group I			
Visual and mechanical	All[1/]	3.3, 3.7 and 3.9	4.6.1
Certification	All	3.3 and 3.4.1.2	4.6.1.1 and 4.6.1.2
Gaging[2/]	All	3.3	4.6.1.3
Projectile extraction[2/]	50[3/]	3.4.3.1	4.6.1.4
Group II[2/ 4/ 5/]			
Velocity-pressure	25	3.5.1.1	4.6.2.1
Waterproof velocity	20	3.5.1.2	4.6.2.2
Action time	20	3.5.1.3	4.6.2.3
Accuracy firing	20	3.5.1.4	4.6.2.4
Function and casualty[6/]	300	3.5.1.5	4.6.2.5 and 4.6.2.5.1
Group III[2/ 4/]			
High pressure (HPT)	25	3.5.2	4.6.2.6
Low pressure (LPT)	25	3.5.3	4.6.2.6
Safety (HEI, API, APT)[7/]	20	3.6	4.6.3
Drop sensitivity (HEI, API)	10	3.6.1	4.6.3.1
Incendiary performance (API)	20	3.5.4	4.6.2.7
Tracer functioning (APT)	20	3.5.5	4.6.2.8
High explosive incendiary (HEI)	20	3.5.6	4.6.2.9
Group IV			
Preparation for delivery		Section 5	4.7

[1/] "All" means that all cartridges covered by this specification will be subjected to this test. The designation of a particular type (API, HEI, etc.) in Group III means that the test is for that type of cartridge only.

[2/] Test not required for FCT cartridge (DL 2051583).

[3/] Quantity of 50 cartridges is for first article tests only, use Table III for quantity of cartridges for quality conformance tests (except HPT and LPT).

[4/] Test not required for dummy cartridge (DL 2605723).

[5/] Not required for HPT and LPT cartridges.

[6/] Test sample shall be equally divided for firing in both MARK 11 and MARK 12 guns.

[7/] Assembled projectiles only to be furnished in this sample. Additional components will be supplied by the testing activity.

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4.5.2 Individual inspections. The individual inspections of Table II, Group I, shall be performed prior to the sampling tests. Failure to meet the specified requirements shall be cause for rejecting any cartridges from the lot.

4.5.3 Lot sampling. Upon completion of the individual inspections and the projectile extraction tests, 500 sample cartridges shall be randomly selected from the inspection lot (using the sampling methods of MIL-STD-105) for ballistic tests in accordance with AD-1324. Failure of any sample to meet the requirements specified herein shall be cause for lot rejection.

4.5.4 Control cartridges. In addition to the sample specified in 4.5.3, the contractor shall deliver from the same component lots that comprise the cartridge lot being tested 40 cartridge cases and primers, 40 projectiles, and sufficient propellant for 40 control cartridges (see 6.2.1) to the testing activity with each lot of cartridges delivered.

4.6 Inspection methods.

4.6.1 Visual and mechanical. The cartridge shall be visually and mechanically inspected for conformance to the drawings (see 3.3), identification and marking (see 3.7), and workmanship (see 3.9).

4.6.1.1 Certification. The contractor shall have available for the Government representative written certification accompanied by objective quality evidence (as defined in MIL-STD-109) that the components used as a part of the cartridge meet the requirements specified herein.

4.6.1.2 Primer resistance. The electrical resistance between the primer contact button and the cartridge case of each primed cartridge case shall be measured, using an approved resistance measuring instrument (with maximum limits of 5.0 milliamperes and 7.5 millivolts) after the crimping operation and prior to waterproofing and propellant loading. Primer resistance shall be certified by the contractor and shall meet the requirements of 3.4.1.2.

4.6.1.3 Gaging. Each cartridge shall be mechanically inspected with the assembled ammunition gage (see 3.8). Failure to enter the gage, within specified tolerances, shall be cause for rejecting any cartridge from the lot.

4.6.1.4 Projectile extraction. Fifty sample cartridges shall be randomly selected from the inspection lot and tested in an approved tensile strength machine. The machine shall apply force at a controlled rate of 0.125 to 0.250 inch per minute of test head travel. The force required to extract the projectile from the crimped cartridge case shall meet the requirements of 3.4.3.1 in accordance with the double sampling plan defined in MIL-STD-105 and shown in Table III.

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TABLE III. Projectile extraction sample.

Sample	Sample size	Cumulative size	Accept defectives	Reject defectives
1st	50	50	0	2
2nd	100	150	1	2

4.6.2 Ballistic tests. Sample Cartridges shall be fired as specified in the detailed test and range procedures of AD-1324.

4.6.2.1 Velocity-pressure. Twenty-five sample cartridges and 10 control cartridges (see 4.5.4) shall be temperature conditioned from 68°F to 72°F for a minimum of 2 hours and tested as follows:

(a) The first five sample cartridges fired from the test barrel shall be considered warming cartridges.

(b) Ten control cartridges loaded by the testing activity shall be interspersed with the last 20 sample cartridges.

(c) The remaining 20 sample cartridges and the 10 interspersed control cartridges will then be fired from the pressure barrel (Drawings SK 679810 or SK 679811).

(d) Peak chamber pressures of the sample cartridges, as determined from the tarage table of the copper crusher cylinder lot or piezoelectric pressure transducers, will be averaged as groups for use in computing the average corrected peak pressure. (The 20 sample cartridges as one group and 10 control cartridges as another group.)

(e) Average pressure for the sample cartridges will be corrected for condition of the barrel by the following formula:

$$\bar{P} = P_1 + (P_2 - P_3)$$

where:

$\bar{P}$ = Corrected peak pressure of the sample cartridges.

P_1 = Average peak pressures measured for the 20 sample cartridges.

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P_2 = Assessed pressure of the control cartridges.

P_3 = Average pressure of the 10 control cartridges measured in the pressure barrel.

Average corrected peak pressure and average corrected velocity shall meet the requirements of 3.5.1.1. Average corrected velocity shall be computed as follows:

$$\bar{V} = V_1 + (V_2 - V_3)$$

where:

$\bar{V}$ = The average corrected velocity for the sample cartridges.

V_1 = Average velocity measured from the 20 sample cartridges.

V_2 = Assigned velocity for the propellant used in the control cartridges.

V_3 = Average velocity measured from the 10 control cartridges.

4.6.2.2 Waterproof velocity. Twenty sample cartridges shall be submerged in water at 68° to 72° for not less than 24 hours. Upon removal from the water, the sample cartridges and 10 control cartridges shall be fired from the same test weapon. The average corrected velocity shall be computed using the formula specified in 4.6.2.1 and shall meet the requirements of 3.5.l.2.

4.6.2.3 Action time. Twenty sample cartridges shall be submerged in water for 24 hours as specified in 4.6.2.2. Upon removal from the water, the sample cartridges shall be single-action fired from the test weapon. The average cartridge action time shall be determined for the cartridges. Average and individual cartridge action times shall meet the requirements of 3.5.l.3.

NOTE: This test maybe performed concurrently with 4.6.2.2.

4.6.2.4 Accuracy firing. Two groups of l0 cartridges each shall be fired to determine the average dispersion from the test weapon at a vertical target placed at a distance of 500 yards. Three warming rounds shall be fired prior to the test. Each group of 10 cartridges shall then be fired, at a uniform rate, to determine

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the vertical and horizontal impact patterns on the target. The average dispersion pattern shall be computed as follows:

$$\bar{D} = \frac{D_1 + D_2 + D_3 + D_4}{4}$$

where:

$\bar{D}$ = Average dispersion pattern in inches.

D_1 = The longest vertical distance in inches between any 2 impacts from the first group of 10 cartridges fired.

D_2 = The longest horizontal distance in inches between any 2 impacts from the first group of 10 cartridges fired.

D_3 = The longest vertical distance in inches between any 2 impacts from the second group of 10 cartridges fired.

D_4 = The longest horizontal distance in inches between any 2 impacts from the second group of 10 cartridges fired.

All measurements shall be made from the center of the impact holes. The average dispersion pattern (D) shall be converted to mils as follows:

$$M = \frac{\bar{D}}{d} \times 1000$$

where:

M = The average dispersion in mils.

$\bar{D}$ = Average dispersion pattern in inches.

d = The distance in inches from gun muzzle to target.

The average dispersion shall be computed and shall meet the requirements of 3.5.1.4.

4.6.2.5 Function and casualty. One hundred fifty sample cartridges shall be fired fully automatic at ambient temperature, in the MARK 4 MOD 0 gun pod. Suitable screens must be located as shown in AD-1324 within 200 feet of the gun muzzle to detect any metal parts separation or breakup of the projectile assembly during flight. The requirements of 3.5.1.5 shall be met.

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4.6.2.5.1 Temperature extremes. The 150 sample cartridges shall be divided into 3 groups and temperature-conditioned to -65°F, ambient, and +165°F for a minimum of 2 hours. The cartridges shall be fired from a MARK 12 gun. Suitable screens must be located as shown in AD-1324 within 200 feet of the gun muzzle to detect any metal parts separation or breakup on the projectile assembly during flight. Cartridges shall meet the requirements of 3.5.1.5.

4.6.2.6 HPT and LPT pressure firing. Sample cartridges shall be fired as specified in the detailed test and range procedures of AD-1324. Twenty-five sample and 10 control cartridges (see 4.5.4) shall be temperature conditioned from 68°F to 72°F for a minimum of 2 hours and tested as Follows:

(a) The first five sample cartridges fired from a pressure test barrel shall be considered warming rounds.

(b) Ten control cartridges loaded by the testing activity shall be interspersed with the last 20 test cartridges.

(c) The 20 cartridges and the 10 control cartridges shall be fired in the pressure test barrel.

(d) Peak pressures of the 20 sample cartridges and the 10 interspersed control cartridges, as determined from the tarage table of the copper crusher cylinder lot or piezoelectric pressure transducers, shall be averaged as groups for use in computing the average corrected pressure. (The 20-sample cartridges are considered as one group and the 10 control cartridges as another.)

(e) Average pressure for the sample cartridges shall be corrected for the condition of the barrel by the following formula:

$$\bar{P} = P_1 + (P_2 - P_3)$$

where:

$\bar{P}$ = Average corrected peak pressure of the sample cartridges.

P_1 = Average peak pressures measured for the 20 sample cartridges

P_2 = Assigned pressure of the 10 control cartridges

P_3 = Average measured pressure of 10 control cartridges in the pressure barrel.

Sample cartridges shall meet the requirements of 3.5.2 or 3.5.3, as applicable.

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4.6.2.7 Incendiary performance. Twenty API cartridges shall be randomly selected from the inspection lot and submerged in water at 68°F to 72°F for 24 hours prior to firing. The projectile shall impact an aluminum plate (Al 2024-T3, QQ-A-250/4) 0.090 inch thick (nominal), positioned normal to the line of fire, located at 200 ±25 yards from the gun muzzle. Acceptance criteria shall be based on photographic evidence that 19 of 20 projectiles fired meet the requirements of 3.5.4 as determined in accordance with AD-1324.

4.6.2.8 Tracer function. Twenty APT cartridges shall be randomly selected from the inspection lot and submerged in water at 68°F to 72°F for 24 hours prior to firing. Visual observation of tracer characteristic shall be made from a suitable location near the gun. Evidence of separation of either the sealant, the entire tracer, or any fragments thereof as determined in accordance with AD-1324 shall cause lot rejection. Burn time requirements of 3.5.5 shall be met on 17 of 20 test projectiles.

4.6.2.9 High explosive incendiary. Twenty HEI cartridges shall be randomly selected from the inspection lot and submerged in water at 68°F to 72°F for 24 hours prior to firing. The projectile shall impact a 0.040-inch (nominal) thick aluminum plate (A1 2024-T3, QQ-A-250/4), positioned normal to the line of fire, located at 200 ±25 yards from the gun muzzle. Acceptance criteria shall be based on photographic evidence that 19 of 20 projectiles fired meet the requirements of 3.5.6 as determined in accordance with AD-1324.

4.6.3 Safety. Twenty safety cartridges each (API, APT, or HEI) shall be loaded with the type and quantity of propellant specified by the testing activity, to provide a chamber pressure of 68,000 ±2,000 psi copper at 70°F. The safety cartridges shall meet the requirements of 3.6 when tested in accordance with AD-1324.

4.6.3.1 Drop sensitivity. Sample HEI or API cartridges shall be dropped from a height of 40 feet and impact nose-down normal to a steel plate or at a velocity of 52 ±5 ft/sec. The sample cartridges shall meet the requirements of 3.6.1 when tested in accordance with AD-1324. Evidence of function shall be cause for lot rejection.

4.7 Preparation for delivery. The packaging, packing and marking shall be inspected to ensure compliance with the requirements of Section 5 prior to shipment.

5. PACKAGING.

5.1 Preservation, packaging, and packing.

5.1.1 Level A. Cartridges shall be preserved, packaged, and packed in small arms ammunition boxes in accordance with LD 256008, 49 CFR Parts 171-178, and CG-108.

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5.2 Marking.

5.2.1 Normal marking. Cartridge containers shall be marked in accordance with MIL-STD-129.

5.2.2 Special marking. In addition to the marking specified in 5.2.1, cartridge containers shall be marked for shipment in accordance with the requirements of OP 2165 (Volume 1), NAVSUP Publication 505/MCO P4030.19, and the Code of Federal Regulations 49 CFR Parts 171-178 and 180-190.

6. NOTES.

6.1 Intended use. The 20MM cartridges described herein are intended for use in aircraft armed with the MARK 4 MOD 0 gun pod and MARK 12 gun.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) MARK and MOD cartridge specified (see 3.1).

(c) When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations (DAR). The first article should be a preproduction sample consisting of l,250 cartridges as specified in 3.2. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

(d) Type and quantity of propellant for HPT and LPT cartridges (see 3.5.2 and 3.5.3).

(e) Government-loaned property (see 3.8).

(f) Responsibility for testing and test facilities if different from 4.1.

(g) Firing inspections (tests) (see 4.3).

(h) Lot identification (see 4.5.1.2).

(i) Control cartridge provisions (see 4.5.4).

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6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the DAR, the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated in the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited as follows:

Paragraph	Data Requirement	Applicable DID
3.10	Ammunition Data Cards	DI-L-1410

(Copies of DIDs required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Combining tests. With approval of the procuring activity, firing tests may be performed concurrently on the same sample cartridges provided that the test results are not affected. The testing facility should make maximum use of this procedure to minimize testing costs.

6.4 Safety precautions. The loading, assembly, and handling of the explosives, subassemblies, and the finished items covered by this specification involve hazardous operations and, therefore, require suitable explosive safety precautions. Standard safety precautions for explosive-loaded items are contained in DoD 4145.26M and are recommended for use by the contractor in loading and assembling the items furnished under this specification.

6.5 Definitions.

6.5.1 Premature burst. Function of an HEI projectile in the gun or in-flight before impact on target.

6.5.2 Premature ignition. Function of an API projectile in the gun or in-flight before impact on target.

6.5.3 Action time. The action time (in milliseconds) is read from the close of the firing key to the projectile's emergence from the barrel.

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6.5.4 Control cartridges. Control cartridges are hand-loaded by the testing activity to establish a propellant charge weight that yields an average velocity within 10 ft/sec of the basic velocity prescribed in the item specification.

6.6 Supersession data. This specification includes the requirements of MIL-C-18835C dated 19 August 1968 and purchase description AS-2583 A dated 12 August 1976.

Preparing activity:
NAVY - AS

Project number: 1305-N835

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MIL-C-85122(AS) Cartridges, 20 MM MARK 100 Series

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☐ VENDOR ☐ USER ☐ MANUFACTURER

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